

Problem 1

(A)

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

$$P_A(t) = \det(A - tI)$$

$$= \det \left(\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} - t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)$$

$$= \det \left(\begin{pmatrix} 1-t & 2 \\ 3 & 4-t \end{pmatrix} \right)$$

$$= \det \begin{pmatrix} 1-t & 2 \\ 3 & 4-t \end{pmatrix}$$

$$= (1-t)(4-t) - 6$$

$$= + (t-1)(t-4) - 6$$

$$= t^2 - 5t + 4 - 6$$

$$= t^2 - 5t - 2$$

can't factor. Use quadratic formula

$$P_A(t) = t^2 - 5t - 2$$

Eigenvalues - $P_A(t) = 0$

$$t = \frac{+5 \pm \sqrt{25 - 4(-2)}}{2}$$

$$= \left(\frac{5 + \sqrt{33}}{2}, \frac{5 - \sqrt{33}}{2} \right)$$

Eigenspaces: - Kernel of $(A - \lambda I_2)$

"the set of all vectors that the matrix sends -2-
in the same direction it was inputted in"

$$\lambda = \frac{5 + \sqrt{33}}{2} \Rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} -3 + \sqrt{33} & 2 \\ 3 & 3 - \sqrt{33} \end{pmatrix} = A'_1$$

$$A'_1 \begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad \frac{-3 + \sqrt{33}}{2}x + 2y = 0 \quad 4y = (3 - \sqrt{33})x \quad y = \frac{3 - \sqrt{33}}{4}x$$

$$v_1 = \begin{pmatrix} 1 \\ \frac{3 - \sqrt{33}}{4} \end{pmatrix}, \quad \text{Ker}(A - \lambda I_2) = \text{span} \left\{ \begin{pmatrix} 1 \\ \frac{3 - \sqrt{33}}{4} \end{pmatrix} \right\}$$

$$\lambda = \frac{5 - \sqrt{33}}{2} \Rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} -3 - \sqrt{33} & 2 \\ 3 & 3 - \sqrt{33} \end{pmatrix} = A'_2$$

$$\frac{-3 - \sqrt{33}}{2}x + 2y = 0$$

$$\frac{3 + \sqrt{33}}{2}x = 2y$$

$$y = \frac{3 + \sqrt{33}}{4}x$$

$$v = \begin{pmatrix} 1 \\ \frac{3 + \sqrt{33}}{4} \end{pmatrix}$$

$$\text{Ker}(A - \lambda I_2) = \text{span} \left\{ \begin{pmatrix} 1 \\ \frac{3 + \sqrt{33}}{4} \end{pmatrix} \right\}$$

Eigenspaces: $\text{span} \left\{ \begin{pmatrix} 1 \\ \frac{3 - \sqrt{33}}{4} \end{pmatrix} \right\}$ and $\text{span} \left\{ \begin{pmatrix} 1 \\ \frac{3 + \sqrt{33}}{4} \end{pmatrix} \right\}$

Problem 1

(B)

$$B = \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix} : a \in \mathbb{R}$$

$$p_B(t) = \det(B - tI_2)$$

$$a - t = 0$$

$$t = a$$

$$= \det \left(\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix} - \begin{pmatrix} t & 0 \\ 0 & t \end{pmatrix} \right)$$

$$= \det \begin{pmatrix} a-t & 1 \\ 0 & a-t \end{pmatrix}$$

Eigenvalues are $a \in \mathbb{R}$

$$= (a-t)^2$$

Eigenspace: $\ker(B - \lambda I_2)$

$$\ker \begin{pmatrix} a-a & 1 \\ 0 & a-a \end{pmatrix} =$$

$$\ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} y=0 \\ x \in \mathbb{R} \end{matrix}$$

$$v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

One eigenvalue exists, $\lambda = a$, multiplicity 2. The eigenspace spans $y=0$, $x \in \mathbb{R}$

(C)

B is diagonalizable if, only, if: multiplicity $\lambda = \dim \ker(A - \lambda I)$

We only have one eigenvalue: $\lambda = a$, multiplicity 2

$$\ker(\dots) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}, \quad \dim(\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}) = 1.$$

$1 \neq 2 \Rightarrow B$ is not diagonalizable.

No value of a works, since we're only concerned with the multiplicity of the eigenvalue, and the kernel is the same for all values of a .

B is not diagonalizable
for any $a \in \mathbb{R}$

Problem 2 $(1-t)^2(-2-t)$

(A)

$$P_A(t) = (t-1)^2(t+2)$$

$$\det \begin{pmatrix} t-1 & 0 & 6 \\ 0 & t+2 & 0 \\ 0 & 0 & t-1 \end{pmatrix}$$

$$\lambda = 1 \quad (t^2 - 2t + 1)(t+2)$$

$$\lambda = -2$$

$$t^3 - 2t^2 + t + 2t^2 - 4t + 2$$

$$t^3 - 3t + 2$$

$$-2 \quad -1$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$\lambda_1 = 1, \ker(A - I_3)$$

$$\ker \begin{pmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

$$\begin{aligned} 6x + 0y + 0z &= 0 \\ 0x + 0y + 0z &= 0 \\ 0x + 0y + (-3)z &= 0 \end{aligned}$$

$$x \in \mathbb{R}, y \in \mathbb{R}, z = 0$$

$$\lambda = 1 \text{ has multiplicity } 2, \dim(\ker(A - I_3)) = 2.$$

$$= \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\lambda_2 = -2, \ker(A + 2I_3)$$

$$\ker \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\lambda = -2 \text{ has multiplicity } 1, \dim(\ker(A + 2I_3)) = 1.$$

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \quad P_B = (t-1)^2(t+2)$$

$$\text{But, } \lambda_1 = 1 \Rightarrow \ker(A - I_3) = \ker \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{pmatrix} \Rightarrow \begin{aligned} y &= 0 \\ x &\in \mathbb{R} \\ z &= 0 \end{aligned}$$

$$\text{multiplicity } \lambda_1 = 2 \text{ but } \dim \ker(A - \lambda_1 I_3) = 1.$$

$$\text{Hence, since } 2 \neq 1, \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \text{ is not diagonalizable.}$$

$$\text{Hence, } \begin{aligned} \bullet A_1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \\ \bullet A_2 &= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \end{aligned}$$

Both have characteristic polynomial $P_A = P_B = (t-1)^2(t+2)$ but only A_1 is diagonalizable.